THE PREIMAGE OF A SCHEME THEORETICALLY DENSE OPEN UNDER A FLAT MORPHISM IS SCHEME THEORETICALLY DENSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof names two characterizations and then the scheme case; the quasi-compactness of V -> Y is part of the counted statement span.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Characterization of scheme theoretically dense open subspaces. Lemma lemma-scheme-theoretically-dense, used at proof lines 6272-6273. Turns the claim into a statement about injectivity of restriction maps.
- Flatness characterized through etale coverings. Lemma lemma-flat-local, used at proof lines 6274-6275. Reduces flatness of f to the scheme case.
- The scheme case for the preimage of a scheme theoretically dense open. Morphisms, Lemma morphisms-lemma-flat-morphism-scheme-theoretically-dense-open, used at proof lines 6276-6278. Is the input that closes the argument.

Source reference. Original lemma 03MM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4754–4754. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0833: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10273–10273. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 081H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10217–10217. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a flat morphism of algebraic spaces over S. Let $V \to Y$ be a quasi-compact open immersion. If V is scheme theoretically dense in Y, then $f^{-1}V$ is scheme theoretically dense in X.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Using the characterization of scheme theoretically dense opens in Lemma 0833 and using the characterization of flat morphisms in terms of étale coverings in Lemma 03MM we reduce to the case of schemes which is Morphisms, Lemma 081H. □